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SIEMENS

Preparation Guide of SAP Business Suite®

Teamcenter Gateway for SAP Business Suite Edition

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Preface

This documentation cannot be used as a substitute for consulting advice, because it can never consider the individual business processes and configuration. Despite our best efforts it is probable that some information about functionality and coherence may be incomplete.

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1. Introduction

The Teamcenter Gateway for SAP Business Suite (**T4S**) software solution is a general purpose integration software that provides data and process integration between Teamcenter® by Siemens Product Lifecycle Management Software Inc. and SAP Business Suite® by SAP AG.

T4S provides a wide range of interactive and workflow functions to transfer and synchronize data between Teamcenter and SAP Business Suite®.

2. Supported SAP Products

Please see: [How to obtain the SAP GUI release and patch level information](#)

The latest information on supported products is available in the readme file *Teamcenter_Gateway_for_SAP_11.4.1_README_for_Teamcenter_*.txt*. It can be downloaded from **GTAC** from Teamcenter and TeamcenterRapid Start / Integrations directory.

Caution:

SAP S/4HANA® is not supported by Teamcenter Gateway for SAP Business Suite! Teamcenter Gateway for SAP S/4HANA is offered for this platform.

Please also see <https://launchpad.support.sap.com/#/notes/147519> for supported products.

SAP GUI for Windows Support Deadlines:

- 7.40: Full support up to January 9th, 2018. Restricted support until June 30th, 2018. No extended support is offered
- 7.50: Full support up to April 9th, 2019. Restricted support until September 30th, 2019. No extended support is offered.

3. Software Prerequisites to Run Teamcenter Gateway for SAP Business Suite

3.1 Legal Information

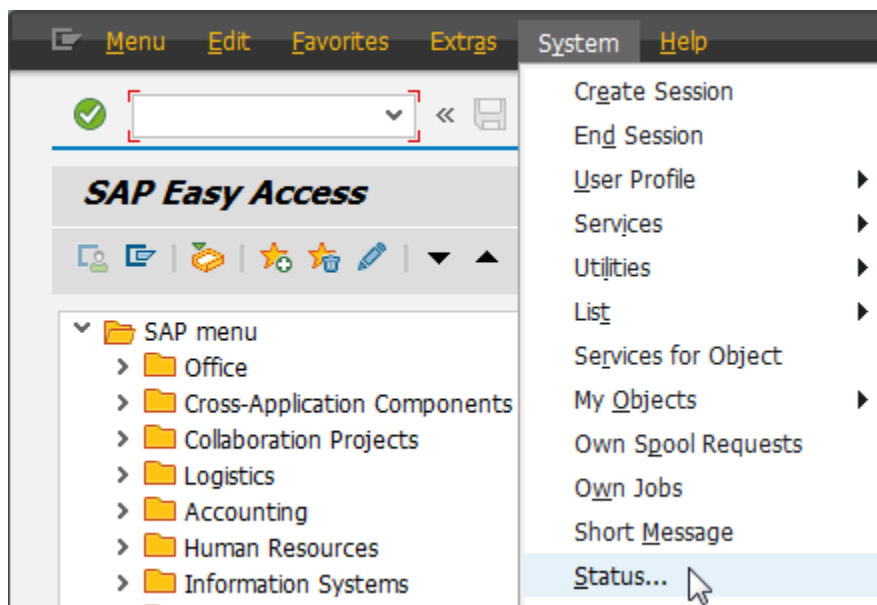
SAP does not allow their software partners to ship files owned by SAP. As a result, some of the files T4S requires to communicate with SAP are not included in the T4S installation package and need to be obtained directly from **SAP ONE Support Launchpad** or with the help of your SAP support partner.

In order to download software from the **SAP ONE Support Launchpad**, a valid web login with user and password is necessary.

3.2 How to Obtain the SAP and SAP GUI Release and Patch Level Information

SAP Version

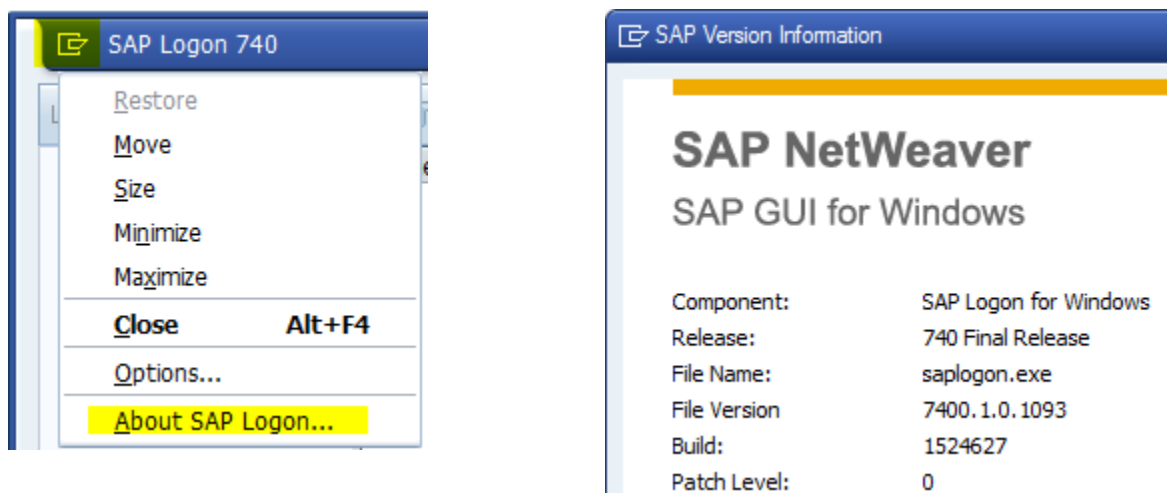
In the main menu point System – Status, check the point "Component version".



System: Status			
Usage data			
Client	800	Previous logon	20.02.2017 11:13:03
User	TPUSER01	Logon	13:38:59
Language	EN	System time	13:50:48
SAP data			
Repository data		SAP System data	
Transaction	SESSION_MANA...	Component version	SAP ECC 6.0
Transaction variant	#####...	Installation number	0020211441
Program (screen)	SAPLSMTR_NAV...	License expiration	31.12.9999
Screen number	100	Unicode System	Yes
Program (GUI)	SAPLSMTR_NAV...		
GUI status	SESSION		
Host data		Database data	
Operating system	Linux	Database system	ORACLE
Machine type	x86_64	Release	10.2.0.2.0
Server name	sapsrv_ET1_20	Name	ET1
Platform ID	390	Host	sapsrv
		Owner	SAPSR3

SAP GUI Version:

In the SAP logon window, click on the icon in the upper left corner (where you may minimize and close windows by default) and select "About SAP logon". Then a new window opens that shows the information. Example from a SAP GUI 7.40 with patch level 0:



3.3 Netweaver RFC SDK

SAP has deprecated the support of RFC SDK by March 2016. Therefore the SAP Netweaver RFC SDK is available as a successor for SAP RFC.

Depending on the SAP system configuration those file versions may be different. Unfortunately T4S cannot check which are the correct files, so in case of a problem with SAP communication (in most cases in DIR handling only) please ask your SAP specialist for help and provide the correct files. The SAP RFC SDK can be downloaded from the [SAP ONE Support Launchpad](#)

The corresponding files have to be copied into the GS bin directory (for example `<GS_ROOT>\bin32` or `<GS_ROOT>\bin64`):

- `libsapnwrfc.dll (.so)`
- `libsapucum.dll (.so)`
- `libicodecnumber.dll (.so)`

For more details, please read `<GS_ROOT>\bin64\readme.sapnwrflib`

Caution:

If the Netweaver runtime library files are missing, the GS will not start.

As the GS can still use some SAP JCO functionalities, it additionally may require the JCO runtime libraries.

If the correct version of the files can't be clearly identified, the best possibility to find the correct package is downloading several packages from SAP and checking if the T4S transaction is successful. In order to avoid a long search, first check the following in the SAP GUI: [How to obtain the SAP GUI](#)

release and patch level information. Then try downloading only the packages from SAP with similar descriptions.

3.4 JCO (Java Connector) Libraries

Using JCO may not be necessary for the basic T4S functionality. Examples where JCO may be necessary are:

- Some T4x web services.
- A couple of not yet migrated standard T4S functions are still using the JCO adapter at the moment, this are in the following areas:
Routing Change
Material Batches
iPPE
Project System
Manufacturing functionality, especially Routing Transfer and Vendor Management.
- Any customer specific functionality using JCO (reading or writing SAP fields that are inaccessible by the default SAP mechanisms).
In previous T4S versions, custom SAP functions were called via JCO. This will still be the case until the code has been replaced with generated code for the NW interface.
- Some T4S features use the SAP JCO adapter without telling it explicitly.

T4S uses SAP JCO Release 3.0.9 (or higher) for all platforms. This version can be downloaded on the SAP Service Marketplace. Depending on the hardware and OS platform (Windows, UNIX, Linux), you need a different JCO package. This Java process is independent from the Java process started with Teamcenter, it may use completely different environment settings.

The system hardware determines the package where to take the SAP JCO library files from. In Linux/UNIX, the command `uname -a` tells some details. As not every hardware is certified with Teamcenter, you may refer to the **hardware and software certifications** to exclude some hardware packages.

Required files:

- sapjco3.dll (.so)
- sapjco3.jar
- sapjco3.pdb

For more details, please see `<GS_ROOT>\bin64\readme.sapnwrflib`.

Caution:

The Java version used in your machines determines whether the 32-bit or 64-bit libraries are used. Depending on the Java version, it might be necessary to use the 32-bit JCO package even on a 64-bit system. In a command shell, enter `java -version`. Check the output: if it does not state anything about 64 bit, it is a 32-bit Java version and needs the 32-bit JCO package. The minimum required Java version for SAP JCO is JSE 1.7.0.

3.5 SAPftp and SAPhttp

T4S may use the SAP tools SAPftp and SAPhttp for doing the check-in of original files to a SAP DIR. The SAPftp and SAPhttp SDK can be downloaded on the [SAP ONE Support Launchpad](#). The corresponding files must be copied into the T4S bin directory `<GS_ROOT>\bin64`. They have to match the ones in used by your SAP system. This can be checked in a command shell as follows:

Example: `C:\temp>C:\PLM\T4S_Apps\bin64\sapftp.exe -V`

```
SAPFTP Unicode
@(#) $Id: //bas/742_REL/src/krn/ftp/ftpmain.c#2 $ SAP
@(#) $Id: //bas/742_REL/src/krn/ftp/ftp.c#2 $ SAP
@(#) $Id: //bas/742_REL/src/krn/ftp/ftpcmd.c#2 $ SAP
SAP release: 742 SAP release no: 7420
```

You may need additional libraries to run the executable that have to be obtained from [SAP ONE Support Launchpad](#):

- `librfc32.dll (.so)` or for unicode `librfc32u.dll (.so)`. Although the RFC interface is no longer supported by SAP, the file transfer still requires these old libraries.
- `icudt34.dll (.so)` or higher
- `icuin34.dll (.so)` or higher
- `icuuc34.dll (.so)` or higher

For more details, please see `<GS_ROOT>\bin64\readme.sapnwrflib`

3.6 SAP GUI for HTML Support

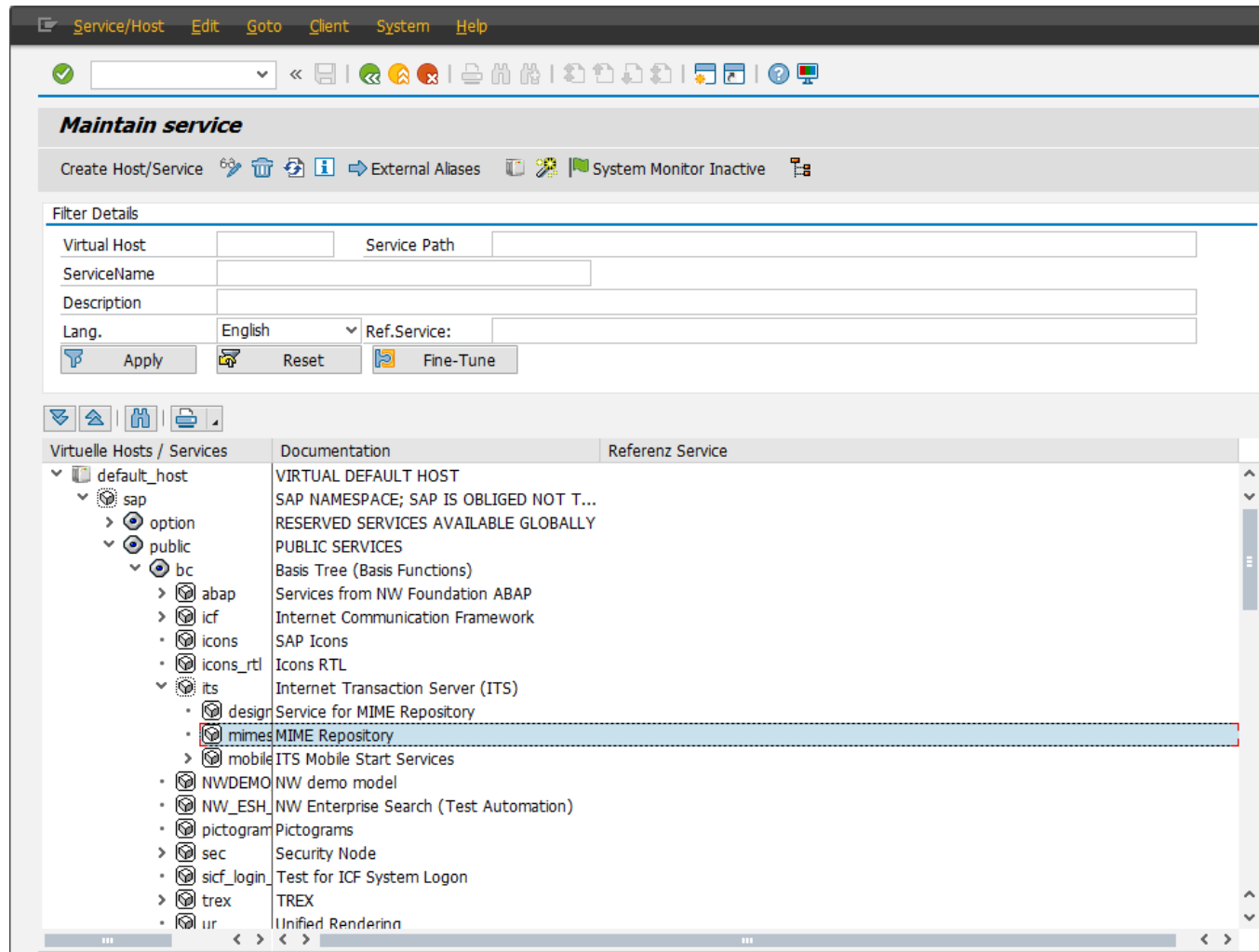
T4S also supports the so called SAP GUI for HTML for a set of display functions (which can be configured via the corresponding T4S Gateway Menu preference). The functionality is based on the Internet Transaction Server (ITS) of the SAP system. More details can be found in the corresponding SAP documentation. This chapter gives an overview of the relevant basic configuration steps.

3.6.1 Services for the SAP GUI for HTML

With the SAP transaction `SICF` it is possible to validate the status of the following services:

- `/sap/public/bc/its/mimes`
- `/sap/bc/gui/sap/its/webgui`

Both have to be activated.



Create/Change a Service

Path: /default_host/sap/public/bc/its/

Service Name: Service (Active)

Lang.: [Other Languages](#)

Description

Description 1	MIME Repository
Description 2	
Description 3	

Service Data | Logon Data | Handler List | Error Pages | Administration

Service Options

☐ Web Service

☐ Do Not Include Inherited Settings

Load Balancing:

SAP Authoriz.:

Session Timeout: (HH:MM:SS)

Compression:

Interactive Options

GUI Link	Not specified	GUI Configuration
Support Accessibility	Not specified	

The screenshot shows the SAP Service Configuration tool interface. At the top, there is a menu bar with 'Service', 'Edit', 'System', and 'Help'. Below the menu bar is a toolbar with various icons. The main window is titled 'Create/Change a Service'. It contains several input fields and buttons:

- Path:** /default_host/sap/bc/gui/sap/its/
- Service Name:** webgui
- Lang.:** English (with a dropdown arrow and an 'Other Languages' button)
- Description:**
 - Description 1: SAP GUI for HTML
 - Description 2: (empty)
 - Description 3: (empty)
- Service (Active):** (checked)

Below these fields are tabs for 'Service Data', 'Logon Data', 'Handler List', 'Error Pages', and 'Administration'. The 'Service Data' tab is selected, showing 'Service Options' and 'Interactive Options'.

Service Options:

- ☐ Web Service
- ☐ Do Not Include Inherited Settings
- Load Balancing:** (empty)
- SAP Authoriz.:** (empty)
- Session Timeout:** 00:00:00 (HH:MM:SS)
- Compression:** Yes (dropdown)

Interactive Options:

- GUI Link:** Yes (dropdown)
- Support Accessibility:** No (dropdown)

There is a 'GUI Configuration' button next to the 'GUI Link' dropdown.

3.6.2 Internet Transaction Server Configuration

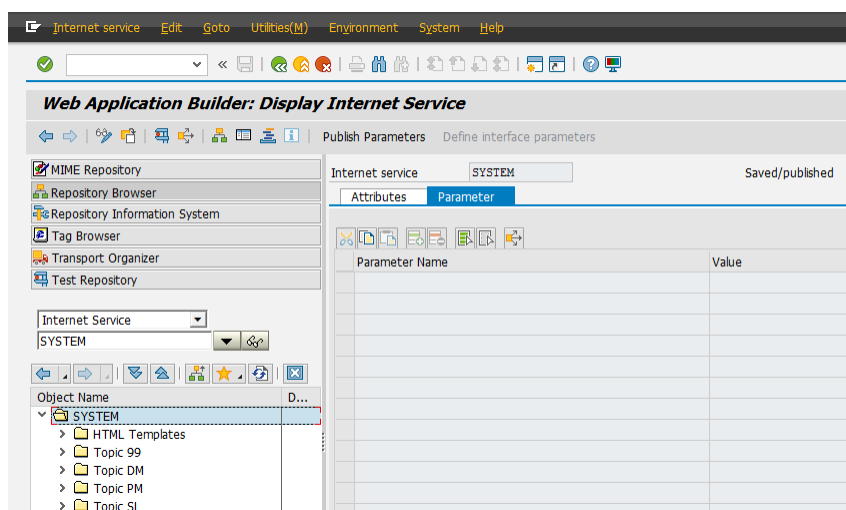
Use transaction SE80 to validate the ITS configuration:

Utilities > Settings > Internet Transaction Server (Tab) > Publish (Tab).

Selecting "Integrated ITS" restricts the publication in the next step to the integrated Internet Transaction Server.

3.6.3 Internet Services SYSTEM and WEBGUI

For the SAP GUI for HTML it is necessary that the "Internet Services" SYSTEM and WEBGUI are published. By this they can be called by a web browser. This status check of the internet services can be managed via the SAP transaction code SE80. If needed the services can be published by the context menu > Publish > Complete Service.



3.7 SAP Transactions via the SAP Portal Support

Beside the T4S support for the SAP GUI for HTML it is also possible to call the corresponding display transactions via the SAP Portal iView URL. This chapter will show the configuration steps that are necessary to enable the feature.

Caution:

The SAP system configuration requires special SAP permissions.

Please ask your SAP service provider for the SAP Portal iView URL which can be used in a web browser.

The following SAP web link shows details how to handle that functionality from SAP directly. It needs a login with a SAP customer account: <http://scn.sap.com/docs/DOC-52102>

For testing purposes, it should be possible to call the SAP Transaction iView directly via the following URL pattern:

```
http://<portal_host>:<portal_port>/irj/portal?NavigationTarget=
ROLES://portal_content/<ContentFolder>/sap_transaction_iview&System=
<ABAP_System_Alias>&TCode=[*]<TCode>
%20<param_name1>=<value1>;<param_name2>=
<value2>&GuiType=<GuiType>
```

The following table shows the definition of the used placeholders (variables) and their corresponding meaning:

Variable	Description
<ContentFolder>	The folder name that was used during the preparation of the iView copy, e.g. TESIS
<ABAP_System_Alias>	The ABAP backend system alias as defined in the portal, e.g. de_tesis_ET2_800
<TCode>	The SAP transaction code. If preceded by a * it will be executed immediately, e.g. *MM03
<param_name1>	The name of the first parameter for the transaction screen (optional), e.g. RMMG1-MATNR
<value1>	The value for the first parameter of the transaction screen, e.g. P-100
<GuiType>	The SAP GUI type to use. This can be either WebGui (SAP GUI for HTML) or WinGui (SAP GUI for Windows). The value is case sensitive!

In this example the transaction MM03 (Material Master Display) is started within the SAP Portal for the material P-100 using the SAP GUI for HTML representation of the transaction:

```
http://sap1.xy.de:55000/irj/portal?NavigationTarget=ROLES://
portal_content/
TESIS/
sap_transaction_iview&System=de_tesis_ET2_800&TCode=*MM03%20RMMG1-MATNR=
P-100&GuiType=WebGui
```

The required T4S configuration for the specific SAP system is described in the **Teamcenter Gateway for SAP Business Suite - Configuration Guide**, chapter *Configure the T4S Gateway Menu Options*.

3.8 SAP OSS Notes (Patches)

Depending on the implemented functionality and use cases, it might be required to implement certain OSS notes in order to fix known in SAP:

- <https://launchpad.support.sap.com/#/notes/>
- <https://support.sap.com/software/patches.html>

The following functionalities are affected:

- Document Info Record File Transfer
- Routing
- Project BOM / WBS BOM / Functional Location BOM

- Characteristic / Product Configurator / Class Sync

Document Info Record File Transfer

The following OSS notes are known to create problems with original file check-in and may have to be removed:

- 1692988
- 1605054

Routing

If released for the target SAP, the following OSS notes must be implemented in order to ensure that the SAP Routing Create and the T4S Routing Change code works as expected:

1481676	EWB: Component assignment creates incorrect PLMZ entries
1734785	EWB: Error during component assignment (change states)
1946879	EWB: The task list sequence is incorrectly saved (reference node)
1388025	Reassigning component assignment to parallel sequence
1915244	Lange Laufzeit oder Dump bei der Synchronisierung (3) (this patch is not mandatory, but recommended)
1966556	RCA: Error caused by SAP Note 1915244 (this note must be installed when note 1915244 is in the system)

The following notes are not required by the Routing Create/Change APIs, but recommended for general use (SAP GUI):

2118418	Inconsistent Component Allocation from CEWB due to incorrect entry in PLZU Table
2106108	Multiple Component Allocations for a single BOM component to the same sequence or parallel sequence.

Project BOM / WBS BOM / Functional Location BOM

If a special BOM type (e.g. Project BOM - also known as WBS BOM, Functional Location BOM or Equipment BOM) is changed by T4S in SAP using a change number, all BOM lines will be referenced with the change number even though they have not been changed.

Download and implement SAP OSS note 1992647: CSAI_BOM_MAINTAIN: All positions are updated with the new change number even though only one position is changed.

Characteristic / Product Configurator / Class Sync

If released for the target SAP, the following OSS notes must be implemented in order to ensure that T4S characteristic transfers (e. g. class sync, product configurator, ...) work as expected:

OSS Note 1684377 (BAPI: Long runtime III): Due to an error in SAP Notes 1640511 and 1604614, the system may not recognize changes to characteristics correctly.

4. SAP Function Calls Used and Permissions Required

4.1 SAP Function Calls Used and Permissions Required by Teamcenter Gateway for SAP Business Suite

Mandatory Functions

The following RFC access permissions are needed in any case:

- function group SYST
- RFCPING
- SYSTEM_RESET_RFC_SERVER
- SCSI_GET_SYSTEM_INFO
- RFC_GET_FUNCTION_INTERFACE

Overview of Functions

A complete overview of SAP functions that could be called can be found in the [T4S API Reference](#) under *List of SAP Function Groups and Functions used by T4S*:



The calls are grouped by permissions.

List of actually Used Function Calls

T4S provides a log file that lists all actually used function calls after they have been used. Ideally you get all SAP permissions for the T4S user(s) during development. Then after development is complete, you can extract a list of functions from the log and give it to your SAP administrator to limit the SAP permissions for the testing phase.

In the BGS Admin GUI you will find the logfile `sys/sap_function_calls.log`. It contains a list off all called SAP functions. Please see *Configure T4x BGS Using BGS Admin GUI* in the [Teamcenter Gateway - Installation Guide](#) on how to login to the Admin GUI. Please use the help function of the Admin GUI for more details.

Caution:

The logfile does not list the following "connector internal" calls:

- RFCPING
- RFC_GET_FUNCTION_INTERFACE
- DDIF_FIELDINFO_GET
- SYSTEM_RESET_RFC_SERVER

5. SAP Transport Packages for Teamcenter Gateway for SAP Business Suite Functionality

In order to implement enhanced SAP ABAP functionality provided by Siemens Industry Software, T4S offers SAP transport packages to add the required additional functionality to SAP.

A complete description of the Transport Packages can be found in the file `<GS_ROOT>\var\template\t4s\sap\TransportPackages\readme.txt`. You will find the transport packages in the directory `<T4x_GS_ROOT>\var\template\t4s\sap\TransportPackages`.

The complete list of Transport Packages is documented in the [T4S API Reference](#):

T4S > List of SAP Function Groups and Functions used by T4S

Caution:

Same as the T4S code itself, the ABAP code T4S provides in the SAP transport packages may change. If you are using such an ABAP please make sure to update it as well when T4S is updated.

In order to check if you already have the latest package, do the following: In both your T4S installations (the new and the old one), go to the directory `<T4x_GS_ROOT>\var\template\t4s\sap\TransportPackages`. Open each zip file that contains an ABAP you implemented in your SAP (e.g. `TESISPLM_T4S_MISC.zip`). Find there one file `K900*.ET1` and one file `R900*.ET1` each (e.g. the two files `K900123.ET1` and `R900123.ET1`). If you find exactly the same file names in the old and the new T4S version, then everything is OK. If the number in the name (representing the ABAP version number of this specific ABAP code), in the new T4S installation is higher than in the old one, you have to update the ABAP in SAP to the new one.

6. Set Up to Trigger Data Transfer From SAP to Teamcenter

6.1 SAP Event Trigger

All transfers of data to Teamcenter will be event triggered only. There is currently no option to provide interactive user actions to initiate data transfer in SAP. Therefore you need to extend available user exits on user actions or else activate BADI in a desired way.

6.2 Z-Table

The Z-Table is a transparent custom table in the SAP database. It is used for cases where data transfer needs to be triggered from within SAP to Teamcenter. Common use case examples:

- An object or a structure is created first in SAP and needs to be transferred to Teamcenter.
- Information on a SAP object is changed and needs to be updated on the corresponding Teamcenter object.

Creating an entry for an object in the table is usually triggered by a SAP event like “save material master data”. On this event a user exit or some other BADI functions catches the object data and can use ABAP functions to create an entry for that object.

The Z-Table and ABAP function to maintain it are provided by Siemens PLM. Use T4S Transport Package *TESISPLM_T4S_ZPTC* to create Z-Table and ABAP functions stored here: `<GS_ROOT>\var\template\t4s\sap\TransportPackages\TESISPLM_T4S_ZPTC.zip`

All the functions and the database table itself are created within the namespace `/TESISPLM/`.

Caution:

Only the `_INT/_INTERN` functions should be used within SAP user exists or BADIs as they will not perform additional RFC authority checks.

6.3 Table Definition /TESISPLM/ZPTC

Table attributes

Description	Production Trigger Cache
Delivery Class	A
Maintenance	Check “Table maintenance allowed”

Table Fields

Fields	Key	Int.	Field Type	Data Type	Length	Dec. places	Check Table
MANDT	X	X	MANDT	CLNT	3	0	T000
ID	X	X	INT4	INT4	10	0	
OBJECT_TYP			CHAR10	CHAR	10	0	
OBJECT_ID			CHAR80	CHAR	80	0	
OBJECT_DESC			CHAR255	CHAR	255	0	
REASON1			CHAR255	CHAR	255	0	
REASON2			CHAR255	CHAR	255	0	
STATUS_E2S			CHAR1	CHAR	1	0	
ERROR_MSG1			CHAR255	CHAR	255	0	
ERROR_MSG2			CHAR255	CHAR	255	0	
ERROR_MSG3			CHAR255	CHAR	255	0	
DELETED			CHAR1	CHAR	1	0	
CREATED_DATE			DATE_	DATS	8	0	
CREATED_TIME			TIME_	TIMS	6	0	
CREATED_USER			ERNAM	CHAR	12	0	
CHANGED_DATE			DATE_	DATS	8	0	
CHANGED_TIME			TIME_	TIMS	6	0	
CHANGED_USER			AENAM	CHAR	12	0	

Table Settings

Data Class	APPL1
Size Category	9
Buffering	Check "Buffering not allowed"

Number Range for Indexes

Caution:

The database table doesn't contain out of the box configured indexes. Adding such indexes is a customization step based on the individual use cases the table should be used for.

To assign IDs automatically on creation of table entries we need to define a number range object in SAP. Please follow these steps:

1. Create the Number Range Object ZPTC (Transaction SNRO)

Number Length Domain: /TESISPLM/ZPTC

Number range transaction: ZPTC

Warning %: 5

Buffer: 1

2. Create an interval (Transaction ZPTC):

Number: 01; From number: 0000000001; To number: 2147483640 (not external)

6.4 Z-Table ABAP Functions

/TESISPLM/ZPTC_INSERT

To insert one line into the table

/TESISPLM/ZPTC_INSERT_INTERN

To insert one line into the table - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_UPDATE

To update one line of the table.

/TESISPLM/ZPTC_UPDATE_INTERN

To update one line of the table - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_DELETE

To delete one line of the table.

/TESISPLM/ZPTC_DELETE_INTERN

To delete one line of the table - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_GET_DETAIL

To read one complete line of the table.

/TESISPLM/ZPTC_GET_DETAIL_INT

To read one complete line of the table - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_DYN_SEARCH

To read the ids of the table which fulfill a given pattern.

/TESISPLM/ZPTC_DYN_SEARCH_INT

To read the ids of the table which fulfill a given pattern - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_DYN_DELETE

To delete the lines of the table which fulfill a given pattern.

/TESISPLM/ZPTC_DYN_DELETE_INT

To delete the lines of the table which fulfill a given pattern - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_DYN_SEARCH

To read the ids of the table which fulfill a given pattern.

/TESISPLM/ZPTC_DYN_SEARCH_INT

To read the IDs of the table which fulfill a given pattern - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

/TESISPLM/ZPTC_DYN_DELETE

To delete the lines of the table which fulfill a given pattern.

/TESISPLM/ZPTC_DYN_DELETE_INT

To delete the lines of the table which fulfill a given pattern - internal, not remote enabled call without RFC authority check - should be used for SAP user exits calls only.

A. Glossary

A

Admin

is the term used in this document for people who install and configure Teamcenter and its components. This is in contrast to the "user" role.

Admin GUI

Teamcenter Gateway Administrative User Interface.

AI-Object

Application-Interface Object

Apps

See "GS".

B

BGS

Basic Gateway Service.

BGS_ROOT

The installation directory of the T4S Basic Gateway Service (e.g. C:\Siemens\T4x_BGS)

BMIDE

Teamcenter Business Modeler IDE (Integrated Development Environment).

BOM

A Bill Of Materials is a list of the parts or components and their quantities that are required to build a product.

D

Data Carrier

A SAP medium on which originals are stored.

Dataview mark-up

is the language understood by the Dataview. The Dataview receives messages written in this language from the T4x server. Such messages can be formatted as XML or JSON. Normally users do not see such messages. They may however appear in log files or error messages. The so called prop mapping (e.g.

t4s_prop_mapping_template.sd) contains TCL commands that compose messages in the Data View mark-up.

DIR

DIR is the abbreviation for a SAP Document Info Record.

Document Key

A Document Info Record is identified by the combination of Document Type, Document Number, Document Part and Document Version.

Document Structure

A Document Structure is like a Bill Of Materials for Documents.

E

EA

stands for Enterprise Application, any software or set of computer programs used by business users to perform various business functions in context of current integration's portfolio with Teamcenter.

ECN

The Engineering Change Notice can also be called an Engineering Change Note, Engineering Change Order (ECO), or just an Engineering Change (EC).

EPM

Enterprise Process Modeling.

G

GRM

The Generic Relationship Management provides a general way in which two objects can be associated via a relationship.

GS

Gateway Service, manages the communication between Teamcenter and the Enterprise Application.

GS_ROOT

The installation directory of the T4S Gateway Service (e.g. *C:\Siemens\T4S_GS* or *C:\UGS\T4S_Apps*)

GUI

Graphical user interface.

I

ITK

The Integration Toolkit (ITK) is a set of software tools provided by Siemens PLM Software that you can use to integrate third-party or user-developed applications with Teamcenter.

J

JDBC

Java Database Connectivity is an application programming interface (API) for the programming language Java, which defines how a client may access a database.

JSON

JavaScript Object Notation is a lightweight data-interchange format¹.

K

KPro

Kpro stands for Knowledge Provider. See also Data Carrier.

L

LOV

List of Values.

M

Mapping

The mapping is part of the T4x configuration. It contains the code that controls the behavior of the data transfer between Teamcenter and the ERP system.

MFK

Multi-key functionality in Teamcenter.

MM

MM is the abbreviation for a SAP Material Master.

1 [JSON.org](https://www.json.org/)

O

Object Link

A relation between SAP objects like Material Master and Document Info Record.

Object Management Record

Belongs to a SAP Change Number and Documents changes of one particular SAP object like a Material Master.

OOTB

Out of the box.

Original

A representation of a file in SAP.

P

PIR

PIR is an abbreviation for a SAP Purchase Info Record.

Portal Transaction

This means that a T4S transfer to SAP that is not triggered by a workflow handler but via the T4S Gateway Menu.

R

RAC

stands for Rich Application Client also referred to as rich client or portal.

Revision Level

Used to show changes with reference to a change to a SAP Material Master or Document Info Record.

S

SAP Portal iView URL

Can be used to show sap content in a browser window.

SSL

Secure Sockets Layer.

T

T4S Gateway Menu

A Teamcenter menu item that is provided by T4S to enable Portal Transactions to SAP and provide SAP connectivity options.

T4x

The entire Teamcenter Gateway product family.

TargetTypeName

This is the T4x internal name for the transaction type. E.g. `MaterialMaster` or `DocumentInfoRecord`.

TC

Teamcenter

TCL

is a high-level, general-purpose, interpreted, dynamic programming language.

TEM

Teamcenter Environment Manager.

Transport Package

A file that contains functions that can be imported to SAP

U

URI

Unified Resource Identifier: a generalized form of a resource locator (URL) and resource name (URN), which just identifies a resource, but is not necessarily sufficient to locate (find) the resource. URIs are often used to identify configurations in Java and other languages. See https://en.wikipedia.org/wiki/Uniform_Resource_Identifier for more details.

URL

Unified Resource Locator: a string with a certain format, allowing to load a resource from a network. URLs are a specific form of URNs.

W

WBS

WBS is an abbreviation for a SAP Work Breakdown Structure.

X

XML

Extensible Markup Language is designed to store and transport data in a format that is both human- and machine-readable.

XRT

stands for XML Rendering Template, also known as XML Rendering Stylesheet. These are XML documents stored in datasets that define how parts of the Teamcenter user interface are rendered. They are used for the Rich Client as well as the Active Workspace.

Z

Z-Table

"Z" is the prefix name for custom tables well-known in SAP world.

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